GRADUATE RESEARCH PLAN STATEMENT
Amphibian immune-metabolic tradeoffs in the face of changing climates

Introduction: The *Batrachochytrium dendrobatidis* (*Bd*) chytrid fungus infects amphibian skin, resulting in the chytridiomycosis disease, which is the leading cause of worldwide amphibian declines and extinctions[1]. Environmental factors such as pH, salinity, and temperature impact how amphibians respond to *Bd*; however, the relationship between temperature and *Bd* susceptibility is not entirely straightforward. The current literature paints a contradictory picture, showing increased *Bd* infection-related mortalities at both warmer and cooler temperatures[2]. *Bd* is a cool-adapted fungus (optimal growth between 15°C and 25°C) so it is not surprising that at warmer temperatures, frogs tend to have lower pathogen loads than those at cooler temperatures. What is surprising is that some research shows that despite relatively low fungal loads, at warmer temperatures frogs may exhibit higher mortalities than frogs at cooler temperatures, bearing greater *Bd* loads. However, other studies indicate that frogs infected at temperatures below those to which they are adapted, likewise readily succumb to *Bd* despite not possessing elevated *Bd* loads[3]. These disparities found in the literature may be explained by the Thermal Mismatch Hypothesis[3], which posits that cool-adapted amphibian species will be more susceptible to *Bd* at warmer temperatures, while warm-adapted species will be more susceptible to this pathogen at cooler temperatures[4]. Notably, poikilothermic vertebrates such as amphibians, which do not regulate their body temperature, face metabolic trade-offs with changes in environmental temperatures, foregoing adaptive immune responses in favor of innate immunity as temperatures decrease[5]. At higher temperatures, amphibians more readily mount adaptive immune responses and exhibit increased lymphocyte numbers[6,7]. However, this increased adaptive immune response has been associated with higher frog mortalities to *Bd* and it is hypothesized that an overactive immune response can be counterproductive in clearing *Bd* infections[7]. Thus, the upper and lower temperature ranges within which amphibians can mount effective innate and adaptive immune responses likely reflect the environmental temperature ranges to which they are adapted. I thus *hypothesize* that **beyond adapted temperatures, increased amphibian susceptibility to *Bd* stems from immune-metabolic tradeoffs combined with off-target immune responses**. In consideration of climate change and the exacerbating amphibian declines, it is imperative that I garner a greater understanding of the interdependence of temperature, *Bd,* and the immune efficacies of these animals. Accordingly, I will use the best-characterized *Xenopus laevis* frog immune model to discern the determinants of the increased amphibian susceptibility to *Bd* in temperature extremes.

My work will be framed by the following two questions:

Objective 1: *Do immune-metabolic tradeoffs contribute to increased Bd susceptibility at low temperatures?* Because immune responses are so metabolically costly and animal metabolisms slow with decreasing temperatures, I hypothesize that *Bd*-infected frog mortalities are at least in part due to limited energy resources being diverted away from other physiological functions and towards the *Bd*-elicited immune responses. To examine this notion, I will rear *X. laevis* froglets (most susceptible to *Bd*) at favorable (19°C) and cold (12°C) conditions before mock or *Bd*-challenge. I will harvest the skins from subsets of these animals at 1 and 3 weeks of infection and subject these to RNA sequencing analyses to discern the immunological states of these tissues. I will also perform immune histology on sections of these skins using monoclonal and polyclonal reagents against macrophage, granulocyte, T cell and B cell markers to determine the skin immune cell compositions and favored response type (innate vs adaptive) under the different temperature and infection conditions. Skin *Bd* loads will be examined concurrently using qPCR assays. The metabolic health of these animals' skin will be assessed in parallel by dispersing sections of the isolated skins into cell suspensions (collagenase-dispersed) and comparing the skin cells from distinct treatments via mitochondrial stress tests, to measure extracellular oxygen consumption rates as readouts for cellular respiration/mitochondrial function. This will demonstrate the effect of low temperature on metabolic function at steady-state and during *Bd* infections.

As alluded to above, frogs infected at low temperatures do not possess elevated *Bd* loads but more readily succumb to infections[3]. This corroborates the idea that upon mounting immune responses, these animals are diverting energy away from homeostatic functions and thus making mortality a greater risk. To test whether this is the case, I will repeat the above experiments, but instead of challenging the frogs with viable *Bd,* I will inject them subcutaneously with heat-killed *Bd* before examining their cellular stress responses. These experiments will untangle how low temperatures impact frog metabolism and immune response, without the confounding effects of a lethal pathogen. I will revisit the key findings of the above studies in the critically endangered Costa Rican harlequin frog, *Atelopus varius*, thereby gaining perspective of these processes in a highly *Bd*-susceptible species.

Objective 2: *How does adaptive immune response change with increasing temperatures and Bd infection?* As described above, poikilothermic vertebrates mount more robust adaptive immune responses with elevated temperatures[5]. Whether these responses clear or exacerbate *Bd* infections is still contentious[7]. To delineate how adaptive immune responses impact frog *Bd* infections at increasing temperatures, I will rear control as well as gamma-irradiated and thus lymphocyte-depleted *X. laevis*[8] at control (19°C) and hot (28°C) conditions and mock-challenge or infect them with *Bd*. At 1 and 3 weeks of infection, the skins of these animals will be examined by RNAseq, histology and qPCR, as above, to determine the immune status, immune cell composition and *Bd* loads within these respective tissues. Concurrently and as above, I will also examine the metabolic health of the skin cells under these conditions. Through these approaches, I will determine whether and which leukocyte populations are overrepresented in healthy and infected skin tissues of animals reared in control and elevated temperatures. I will also identify how these components track with skin-*Bd* loads compared to frogs lacking their lymphoid compartment and thus reduced adaptive immune capacity. Key findings will be recapitulated in *A. varius.*

INTELLECTUAL MERIT

Although there is research on amphibian susceptibility to *Bd* at different temperatures, there is a lack of research on the actual immune processes and metabolic trade-offs underlying this susceptibility. My work will illuminate the molecular processes underlying the Thermal Mismatch Hypothesis, thereby defining the molecular mechanisms behind the impacts of changing climates on the already declining amphibian populations. The results of this work will help conservation efforts already in place to reduce amphibian extinctions. As key findings from this work will be reexamined in *A. varius*; the results may help conservation efforts towards restoring this frog to the wild.

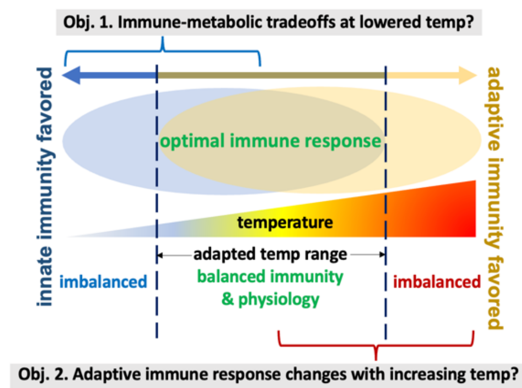

Figure 1. My research objectives, framed by predicted temperature-immune interactions.

BROADER IMPACTS

To increase public awareness of the amphibian declines, I will visit the local School Without Walls highschool and work with the Alice Ferguson Foundation to give guest lectures on climate change and the amphibian declines to junior high and high school students. I will also engage with undergraduate researchers through the Harlan Undergraduate Research inititative, helping to host them in the Grayfer lab, engaging them in the proposed studies and thus simultaneously faustering their appreciation of the importance of basic research, climate change and the global amphibian declines. I will start an initiative to pair high school and undergraduate underrepresented minorities with research opportunities in the DC, Maryland, and Virginia area through GWU and by means of a designated website. I will raise public awareness of climate change and the amphibian declines through this website and via social media.

I sincerely believe that the proposed investigations and outreach activities will enrich the scientific understanding and public awareness of the impacts of climate change on the global amphibian declines and will provide new avenues for slowing and counteracting the amphibian losses.

1. Scheele et al. Science 2019 2. Turner et al. J Wildlife Dis 2021 3. Cohen et al. Ecology Letters 2017 4. Neely et al. Biol Conserv 2020 5. Greenspan et al. Dev Comp Immunol 2017 6. Weerathunga et al. Frontiers Zool 2020 7. Savage et al. Molec Ecol 2020 8. Rollins-Smith, Robert Cold Spring Harbor Prot 2019 9. Weerathunga, Rajapaksa Frontiers Zool 2020.